

MA1521 L1 Week 04 Mon 20260831

Sketch five level curves for each of the following functions.

Also, sketch the portion of the graph of the function lying in the first octant.

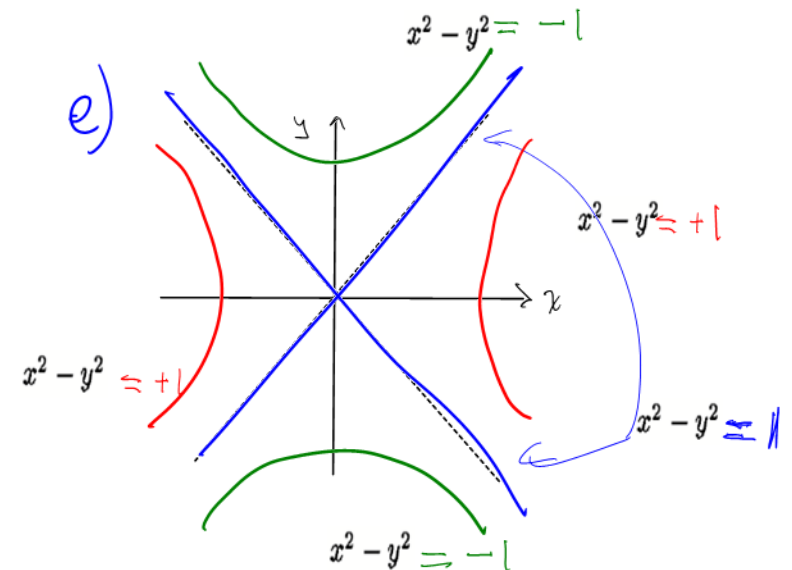
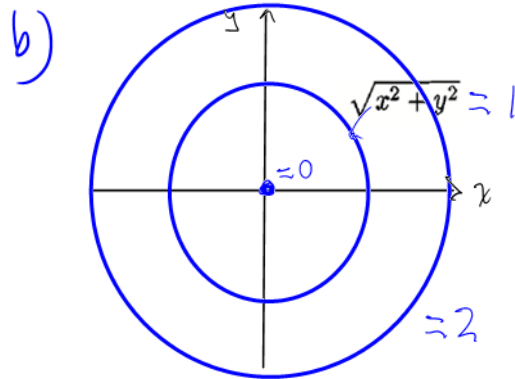
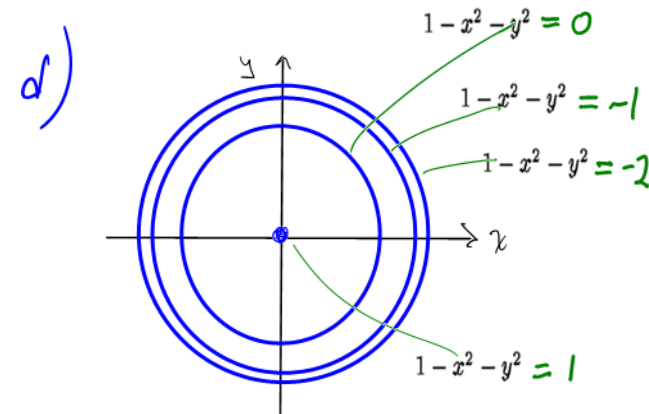
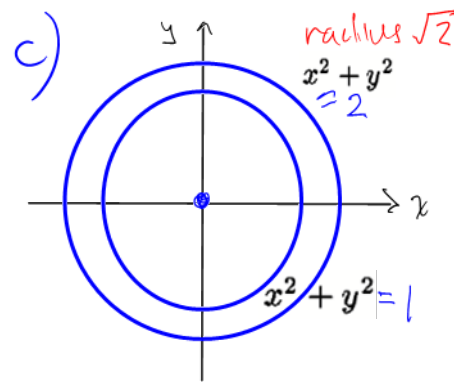
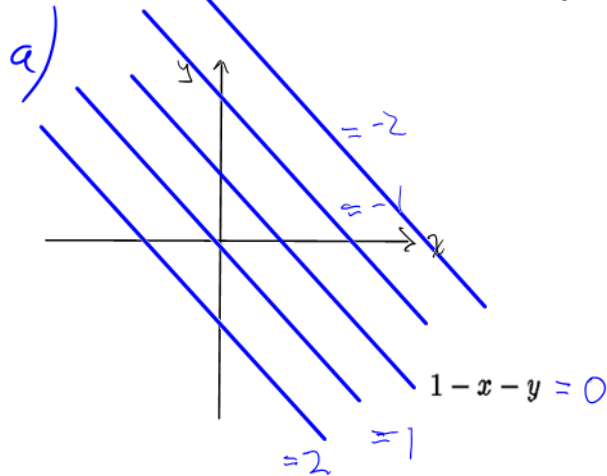
a) $1 - x - y$

b) $\sqrt{x^2 + y^2}$

c) $x^2 + y^2$

d) $1 - x^2 - y^2$

e) $x^2 - y^2$



Calculate the first partial derivatives of each of the following functions:

$$x^3y - 3xy^2 + 2y^2$$

$$\sin(3x + 2y)$$

$$x \ln(2x + y)$$

$$x^2z - 2yz^3$$

$f =$ $f(x, y) =$	$x^3y - 3xy^2 + 2y^2$	$\sin(3x + 2y)$	$x \ln(2x + y)$	$f(x, y, z) = x^2z - 2yz^3$
$\frac{\partial f}{\partial x}(x, y) =$ $f_x(x, y) =$	$3x^2y - 3y^2$	$3\cos(3x + 2y)$	$\ln(2x + y)$ $+ x \cdot \frac{2}{2x + y}$	$f_x(x, y, z) = 2xz$ $f_y(x, y, z) = -2z^3$
$\frac{\partial f}{\partial y}(x, y) =$ $f_y(x, y) =$	$x^3 - 6xy + 4y$	$2\cos(3x + 2y)$	$\frac{x}{2x + y}$	$f_z(x, y, z) = x^2 - 6yz^2$

Equality of mixed partials

$$\left. \begin{aligned} f_{xy} &= (f_x)_y = \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x} \right) \\ f_{yx} &= (f_y)_x = \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial y} \end{aligned} \right\} \text{defn}$$

Verify that $f_{xy} = f_{yx}$ for each of the following:

f = a) $x^m y^n$, (m, n positive integers) b) $\frac{x}{x+y}$ c) $\cos(x^2 + y)$

f_x = $m x^{m-1} y^n$

f_{xy} = $mn x^{m-1} y^{n-1}$

f_y = $n x^m y^{n-1}$

f_{yx} = $mn x^{m-1} y^{n-1}$

$\frac{1(x+y) - x \cdot 1}{(x+y)^2} = \frac{y}{(x+y)^2}$

$\frac{1 \cdot (x+y)^2 - y \cdot 2(x+y)}{(x+y)^4} = \frac{x^2 - y^2}{(x+y)^4} = \frac{x-y}{(x+y)^3}$

$\frac{-x}{(x+y)^2}$

$\frac{-1(x+y)^2 - (-x)2(x+y)}{(x+y)^4} = \frac{x^2 - y^2}{(x+y)^4} = \frac{x-y}{(x+y)^3}$

$-\sin(x^2+y) \cdot (2x) = -2x \sin(x^2+y)$

$-2x \cos(x^2+y)$

$-\sin(x^2+y) \cdot 1$

$-\cos(x^2+y) \cdot 2x$

By using $f_{xy} = f_{yx}$, tell for what value of the constant a there exists a function $f(x, y)$ for which $f_x = axy + 3y^2$, $f_y = x^2 + 6xy$, and then using this value, find such a function by inspection.

Get $f_{xy} = \frac{\partial}{\partial y} (f_x) = \frac{\partial}{\partial y} (axy + 3y^2) = ax + 6y$

$f_{yx} = \frac{\partial}{\partial x} (f_y) = \frac{\partial}{\partial x} (x^2 + 6xy) = 2x + 6y$

By eq. of mixed partials,
must have $a = 2$

Hence $f_x = 2xy + 3y^2$. Have fn $f(x, y)$, such that $\frac{\partial f}{\partial x} = 2xy + 3y^2$
What is $f(x, y)$?

$f(x, y) = \int \frac{\partial f}{\partial x} dx = x^2 y + 3xy^2 + \text{const wrt } x$
i.e. fn of y
say $g(y)$

If $f(x,y) = x^2y + 3xy^2 + g(y)$,

then $f_y(x,y) = x^2 + 6xy + \frac{dg}{dy}(y)$ must equal $x^2 + 6xy$ from hypothesis

hence $\frac{dg}{dy}(y) = 0$, so $g(y)$ is a const. fn of y .

hence $f(x,y) = x^2y + 3xy^2 + c$ where $c \in \mathbb{R}$ is const.

Harmonic fns

Show the following functions $w = f(x, y)$ satisfy the equation $w_{xx} + w_{yy} = 0$ (called the two-dimensional **Laplace equation**):

a) $w = e^{ax} \sin ay$ (a constant)

$$w_x = a e^{ax} \sin ay$$

$$w_{xx} = a^2 e^{ax} \sin ay$$

$$w_y = a e^{ax} \cos ay$$

$$w_{yy} = -a^2 e^{ax} \sin ay$$

$$\begin{aligned} w_{xx} + w_{yy} &= a^2 e^{ax} \sin ay \\ &\quad - a^2 e^{ax} \sin ay \\ &= 0 \end{aligned}$$

b) $w = \ln(x^2 + y^2)$

$$w_x = \frac{2x}{x^2 + y^2}$$

$$w_{xx} = \frac{2(x^2 + y^2) - 2x(2x)}{(x^2 + y^2)^2}$$

$$= 2 \cdot \frac{y^2 - x^2}{(x^2 + y^2)^2}$$

$$w_y = \frac{2y}{x^2 + y^2}$$

$$w_{yy} = \dots = 2 \cdot \frac{x^2 - y^2}{(x^2 + y^2)^2}$$

$$w_{xx} + w_{yy} = 0$$

(x, y, z)

(x, y, w)

Give the equation of the tangent plane to each of these surfaces at the point indicated.

a) $z = xy^2$, $(1, 1, 1)$ b) $w = y^2/x$, $(1, 2, 4)$

Recall: eqn of plane thru (x_0, y_0, z_0) is given by

$$z - z_0 = A(x - x_0) + B(y - y_0)$$

(unless plane is "vertical")

For tangent plane at (x_0, y_0, z_0) on graph of $z = z(x, y)$

must have $A = \frac{\partial z}{\partial x}(x_0, y_0)$ and $B = \frac{\partial z}{\partial y}(x_0, y_0)$

a) $\frac{\partial z}{\partial x} = y^2$ At $(x_0, y_0, z_0) = (1, 1, 1)$, get $A = \frac{\partial z}{\partial x}(x_0, y_0) = 1$ so eqn of tgt plane is
 $\frac{\partial z}{\partial y} = 2xy$ $B = \frac{\partial z}{\partial y}(x_0, y_0) = 2$ $z - 1 = 1(x - 1) + 2(y - 1)$

b) $w_x = -\frac{y^2}{x^2}$ At $(x_0, y_0, z_0) = (1, 2, 4)$, get $A = \frac{\partial w}{\partial x}(x_0, y_0) = -4$ so eqn of tgt plane is
 $w_y = \frac{2y}{x}$ $B = \frac{\partial w}{\partial y}(x_0, y_0) = 4$ $w - 4 = -4(x - 1) + 4(y - 2)$

- a) Find the equation of the tangent plane to the cone $z = \sqrt{x^2 + y^2}$ at the point $P_0 : (x_0, y_0, z_0)$ on the cone.
- b) Write parametric equations for (the ray from the origin passing through P_0) and using them, show the ray lies on both the cone and the tangent plane at P_0 .

a) Want $\frac{\partial z}{\partial x}(x_0, y_0)$, $\frac{\partial z}{\partial y}(x_0, y_0)$

have $z^2 = x^2 + y^2$

Implicit diff. wrt x : $2z \cdot z_x = 2x$

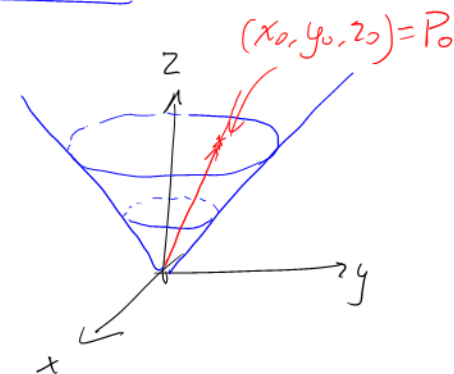
wrt y : $2z \cdot z_y = 2y$

At $P_0 : (x_0, y_0, z_0)$, get $z_x(x_0, y_0) = \frac{2x_0}{2z_0} = \frac{x_0}{z_0}$

$z_y(x_0, y_0) = \frac{2y_0}{2z_0} = \frac{y_0}{z_0}$

$\therefore$ Eqn of tangent plane at $P_0 : (x_0, y_0, z_0)$ is

$$z - z_0 = \frac{x_0}{z_0} (x - x_0) + \frac{y_0}{z_0} (y - y_0)$$



$$(z_0 = \sqrt{x_0^2 + y_0^2})$$

$$\text{i.e. } z_0 z - z_0^2 = x_0 x - x_0^2 + y_0 y - y_0^2$$

$$\text{i.e. } z_0 z = x_0 x + y_0 y \quad \text{--- (B)}$$

b) Param. eqn of ray OP₀ is $\begin{pmatrix} x \\ y \\ z \end{pmatrix} = t \cdot \begin{pmatrix} x_0 \\ y_0 \\ z_0 \end{pmatrix} = \mathbf{r}(t)$

Ray OP₀ lies on cone bec.

eqn A

$$\text{"z-comp of } \mathbf{r}(t)\text{"} = \sqrt{\text{"x-comp of } \mathbf{r}(t)\text{"}^2 + \text{"y-comp of } \mathbf{r}(t)\text{"}^2}$$

i.e. $t z_0 = \sqrt{(t x_0)^2 + (t y_0)^2} \quad \leftarrow \text{substitutes eqn (A)}$

Ray OP₀ lies on tgt plane at P₀ bec.

$$z_0(t z_0) \stackrel{?}{=} x_0(t x_0) + y_0(t y_0) \quad \leftarrow \text{substitutes eqn (B)}$$